

04 - Vaginal Progesterone Before Twenty-Four Weeks

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

Vaginal progesterone has its strongest evidence in one kind of patient: **the woman carrying a single baby, feeling perfectly well, whose cervix is found to be short on a mid-trimester scan.** This lecture builds that claim from the trials, and it marks, just as carefully, the point where the evidence runs out.

What a Short Cervix Actually Is

It helps to begin with what a short cervix actually is, because that turns out to decide everything that follows. The cervix stays long and firm because its collagen is woven into a tight, cross-linked mesh. For the cervix to shorten, that mesh has to loosen and break apart, and several quite different processes can bring that about. Sometimes the **local progesterone signal weakens**, and the mesh softens of its own accord. Sometimes there is **infection inside the uterus**, and the inflammation switches on enzymes that digest the collagen directly. Sometimes there is **bleeding behind the membranes**. And sometimes it is plain **mechanical load**, the kind the uterus feels when it is stretched by twins. Four different drivers, and each one of them arrives at the very same place: a short cervix on the ultrasound. The scan shows you the shortening. What it cannot show you is which of those processes produced it. And that matters enormously, because progesterone can only reach one of them — the cervix that shortened because its own progesterone signaling was failing. For the cervix that shortened from infection, or from bleeding, or from sheer stretch, replacing progesterone does very little.

Fonseca: The Effect at the Extreme

With that in hand, the trials become easy to read. Start with **Fonseca, in two thousand seven**. Fonseca went after the most extreme version of the problem, enrolling women whose cervix had already shortened to **fifteen millimeters or less**, and gave them progesterone vaginally, every night. The effect was large. Delivery before thirty-four weeks fell from **about a third of these women to about a fifth**. One in three became one in five. That is a real benefit in a genuinely high-risk group, and it raises an obvious next question: does the same benefit survive when the cervix is short, but not quite so dramatically short?

Pregnant: Does It Hold at a Less Extreme Cervix?

That is precisely the question the **Pregnant trial** took up a few years later. Pregnant enrolled women one step less extreme, with a cervix between **ten and twenty millimeters**, and treated them with a daily vaginal gel. It found the same halving: birth before thirty-three weeks fell from **about sixteen percent to about nine percent**. So now two trials, sitting at two different cervical lengths, point the same way with the same strength. When studies that enrolled noticeably different patients agree that closely, the effect begins to look like something you can trust.

The Pooled Evidence

The next question is whether that agreement survives being pooled properly, and here the individual-patient meta-analyses come in — **Romero and colleagues**, gathering the raw data from every good short-cervix trial. Across women with a cervix of **twenty-five millimeters or less**, the reduction in early birth held: birth before thirty-three weeks came down by **nearly forty percent**. But the pooled data did something the single trials could only hint at. They showed the babies did better, not only the birth statistics — less respiratory distress, less of the combined burden of newborn illness and death. The benefit reached past the delivery date and into the nursery.

When "Benefit" Means Time, Not a Threshold

And they let us sharpen what the word benefit even means. The updated meta-analysis found that progesterone pushed the average gestational age at delivery up by about **three-quarters of a week — roughly five days**. Five days sounds slight if you hear it flatly. It stops sounding slight once you picture those five days moving a baby out of the window where being born early does the most damage. A drug can help without ever changing whether a birth counts as preterm on a chart, simply by buying time at the part of pregnancy where time is worth the most. That idea will matter for every trial ahead, because some of them measure a threshold and some of them measure time, and the two do not always move together.

OPPTIMUM: Drawing the Boundary

Then comes the trial that seems, at first, to break the pattern. **Opptimum** opened its doors much wider. Rather than selecting women by a short cervix, it gathered a broad high-risk group — some chosen because of a prior early birth, some by other markers — and gave them the same vaginal progesterone. And the benefit largely disappeared. Its obstetric outcome barely moved, the newborn outcomes did not improve, and even

the children's development at two years looked the same. The temptation is to see a contradiction, but the disappearance is exactly what the biology predicts. Progesterone helps the cervix that shortened from failing progesterone signaling. When a trial fills up with women chosen by history instead of by a short cervix, it dilutes those responders with women whose risk runs through infection, or inflammation, or pathways progesterone never touches. Spread one drug across all of them, and the average effect thins to nothing. So Opptimum earns its place by drawing the boundary: it shows the benefit lives wherever the cervix is genuinely short, and fades wherever the enrollment drifts away from it.

The Regimen — and Why Vaginal Is Not 17-OHPC

That gives us enough to be specific about the treatment itself. The regimen from the strongest trials is progesterone given vaginally — **two hundred milligrams as a nightly capsule** in Fonseca, **ninety milligrams as a daily gel** in Pregnant — started in the mid-trimester and carried through to around thirty-four to thirty-six weeks. Every part of that has a reason. **Vaginal**, because the drug needs to act locally, on the cervix and the uterine lining where the biology is happening, rather than circulate through the whole body. **Nightly**, because that is how the trials dosed it, and because lying down keeps it where it was placed. **Carried into the mid-thirties of weeks**, because that is how long the trials kept treating, across the window where an early birth is most dangerous.

One distinction protects you from a common error. This is vaginal, natural progesterone, the same molecule the body already makes. **Seventeen-hydroxyprogesterone caproate** is a different thing entirely: a synthetic compound, given as a weekly injection into muscle. Different molecule, different route, and, as its own lecture will show, a far more troubled evidence base. When someone says progesterone for preterm birth, the first move is always to ask which one, because the two do not share their evidence.

What the Guidelines Do With This

The guidelines take all of this and turn it into a line you can act on. The **Society for Maternal-Fetal Medicine** draws that line for the asymptomatic singleton with no prior early birth, screened before twenty-four weeks. If the cervix measures **twenty millimeters or less**, they recommend vaginal progesterone outright. Between **twenty-one and twenty-five millimeters**, they move to shared decision-making — an offer made alongside an honest conversation, rather than a firm recommendation. That gap between recommend and discuss is the guideline reading how strong the evidence is at each cervical length, and handing you a matching strength of advice.

The **American College** reads the same trials from a slightly different angle. For a woman whose risk comes from a prior spontaneous early birth rather than from a cervix already seen to be short, they lean on watching the cervix over time, and they hold that vaginal progesterone stands on solid evidence for her only once a short cervix appears. Two bodies, one set of trials, and the whole difference between them comes down to which patient is standing in front of you.

The Bedside Rule

Step back, and the trials tell one story. The benefit tracks a single variable — **how tightly each study selected for a genuinely short cervix**. Tighten the enrollment around that cervix and the effect is strong and repeatable; loosen it toward general risk and the effect fades. That is the signature of a drug acting on one specific pathway rather than on preterm birth at large. And it leaves a rule clean enough to carry into any room: **reach for vaginal progesterone when the cervix in front of you is measured short, found early in the mid-trimester, with nothing else competing to explain why it shortened**.

One Patient, and a Caution

Put a real woman in that room. She is twenty-two weeks, carrying one baby, entirely well, and a routine scan reads her cervix at **eighteen millimeters**, with no history of an early birth. Line her up against everything we have built. She is short, she is early, she is asymptomatic, and nothing else is competing to explain the finding. She is, almost exactly, the woman Fonseca and Pregnant enrolled. So the evidence behind offering her progesterone is real, and you can say so without dressing it up.

That last part is worth guarding. The pull, once a drug works somewhere, is to reach for it everywhere — the distant history, the normal cervix, the twin pregnancy, the woman already in labour. The trials keep answering that the benefit lives inside the phenotype and thins outside it. And evidence can fail outright: one of the early, encouraging progesterone studies was later retracted, its data no longer to be trusted. The discipline that protects the patient is the one that has run through this whole lecture. Give the drug to the woman the trials actually studied, and say plainly, everywhere else, how much less we know.